

☐ Database Management Systems (DBMS) - 30 Multiple Choice Questions

Q1. What is a primary key?

- A) A key that allows duplicate values
- B) A unique identifier for a row in a table
- C) A key used for encryption
- D) A foreign key reference

Q2. Which SQL clause is used to filter rows?

- A) GROUP BY
- B) ORDER BY
- C) WHERE
- D) HAVING

Q3. What does ACID stand for in databases?

- A) Atomicity, Consistency, Isolation, Durability
- B) Atomicity, Concurrency, Isolation, Data
- C) Access, Control, Isolation, DBMS
- D) Algorithms, Consistency, Integrity, Data

Q4. Which command is a Data Definition Language (DDL) command?

- A) INSERT
- B) UPDATE
- C) CREATE
- D) SELECT

Q5. What is the default port for PostgreSQL?

- A) 3306
- B) 5432
- C) 8080
- D) 1521

Q6. Which constraint ensures that all values in a column are distinct?

- A) NOT NULL

- B) CHECK
- C) UNIQUE
- D) FOREIGN KEY

Q7. What does the COUNT () function do in SQL?

- A) Returns the sum of values
- B) Returns the average of values
- C) Returns the number of rows
- D) Returns the maximum value

Q8. Which normalization form eliminates partial dependencies?

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF

Q9. What is a Foreign Key?

- A) A key used to connect two tables
- B) A key that must be unique
- C) A key that cannot contain NULL values
- D) A temporary key for queries

Q10. Which SQL join returns all rows when there is a match in either left or right table?

- A) INNER JOIN
- B) LEFT JOIN
- C) RIGHT JOIN
- D) FULL OUTER JOIN

Q11. Which data type is ideal for storing text like a Roll Number or Employee ID?

- A) INTEGER
- B) VARCHAR
- C) BOOLEAN
- D) DATE

Q12. What does DML stand for?

- A) Data Management Language
- B) Data Manipulation Language
- C) Data Definition Language
- D) Digital Market Language

Q13. Which command is used to remove all records from a table without deleting the table structure?

- A) DROP
- B) REMOVE
- C) TRUNCATE
- D) DELETE ALL

Q14. What does the LIKE operator do in SQL?

- A) Compares two numbers
- B) Searches for a specified pattern in a column
- C) Joins two tables
- D) Orders the results alphabetically

Q15. In SQL, what wildcard character represents zero or more characters?

- A) _ (Underscore)
- B) * (Asterisk)
- C) % (Percent)
- D) \$ (Dollar)

Q16. Which clause is used to sort the result-set in SQL?

- A) SORT BY
- B) ARRANGE BY
- C) GROUP BY
- D) ORDER BY

Q17. What is the main purpose of database indexes?

- A) To encrypt database records
- B) To speed up data retrieval
- C) To reduce database storage size

D) To back up database data

Q18. Which command is used to save changes permanently in a transaction?

A) ROLLBACK

B) COMMIT

C) SAVEPOINT

D) UPDATE

Q19. What is a view in SQL?

A) A physical table storing copies of data

B) A virtual table based on the result-set of an SQL statement

C) A backup of a table

D) A database user role

Q20. Which SQL keyword is used to eliminate duplicate rows from the result?

A) UNIQUE

B) DISTINCT

C) DIFFERENT

D) REMOVE

Q21. What happens if you run a DELETE statement without a WHERE clause?

A) The table structure is dropped

B) An error occurs

C) All rows in the table are deleted

D) Only the first row is deleted

Q22. Which aggregate function returns the highest value in a column?

A) HIGHEST()

B) MAX()

C) TOP()

D) SUM()

Q23. What does 1NF (First Normal Form) require?

A) No partial dependencies

B) No transitive dependencies

C) Atomic values (no multi-valued attributes)

D) Presence of a foreign key

Q24. Which clause is used with aggregate functions to filter groups?

A) WHERE

B) HAVING

C) GROUP BY

D) FILTER BY

Q25. What does the **GRANT command do?**

A) Gives user privileges to a database object

B) Revokes user privileges

C) Creates a new user

D) Changes user password

Q26. Which database model represents data in nested tree structures?

A) Relational Model

B) Network Model

C) Hierarchical Model

D) Object-Oriented Model

Q27. What is a subquery?

A) A query that runs after the main query

B) A query nested inside another query

C) A simple index search

D) A query that handles database connections

Q28. What does NoSQL stand for?

A) New SQL

B) Not Only SQL

C) Numbered SQL

D) No Standard Query Language

Q29. Which of the following is a popular NoSQL database?

A) PostgreSQL

B) MySQL

C) MongoDB

D) Oracle

Q30. What does the COALESCE function do?

A) Counts the non-null values

B) Returns the first non-null value in a list

C) Combines two columns

D) Deletes null rows